PROGRESS REPORT

Process Flow Calculations for Ethanol Production Utilizing Wind Turbines
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Report Prepared
By

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At the request of B.E.A.R. (letter dated October 22, 2003), I have performed mass and energy calculations for a process to produce ethanol from water and air, using energy derived from sea-borne wind turbines. The calculations are based on a number of parameters specified by BEAR and also on some assumptions related to the catalytic step to convert CO_2 and H_2 to ethanol. In terms of organization, this report will discuss the input parameters/assumptions, the specific results of the calculations and some alternatives and future efforts that appear to be warranted.

Input Parameters and Assumptions

The starting point for the calculations was based on the availability of 1 MW of power. The overall process was also described by BEAR as consisting of an electrolyzer (to produce hydrogen from water), a distillation unit to separate CO_2 from air and a catalytic reactor to react the hydrogen and CO_2 to produce ethanol, according to the reaction:

$$2CO_2 + 6H_2 \rightarrow C_2H_5OH + 3H_2O$$

It was further specified that the distillation unit would require 0.333 hp to distill 1 SCFM of air, but that 0.25 hp could be recovered from the separated oxygen and nitrogen. Thus, it takes a net power of 0.083 hp to produce CO_2 from 1 SCF of air. With respect to hydrogen generation from the electrolysis of water, operational parameters were taken from the Stuart Energy website (www.stuartenergy.com), which states that their IMET 300 analyzer requires 6.591 KW/mol/min of H_2. Based on a brief literature survey, very little was found relative to catalyst systems that are active to convert CO_2 and H_2 to ethanol. Consequently, the catalytic step was approached in two ways. The first was based on thermodynamic equilibrium calculations (at 800 psig and reactor temperature). However, when both hydrocarbon and alcohol compounds were allowed, equilibrium predicted that the only products were methane and traces of lower hydrocarbons. Under the assumption that alcohols were the only hydrocarbons formed, the predicted CO_2 conversion was 38% and the total predicted alcohol selectivity was 80%. The second approach assumed that the product selectivities were the same as those reported in the one literature article that was found (Inui, T. and T. Yamamoto, Catal. Today, 45 (1998) 209-214); that is, a product selectivity slate that contained 20% ethanol, 5 % methanol, 60% methane and 15% carbon monoxide. In this case the CO_2 conversion was also about 38%.

Results

Figure 1 shows the process flow diagram. Here, “E”, is the electrolyzer, “Cr” is the cryogenic distillation unit, “C” is a compressor to bring the H_2 and CO_2 up to reactor

pressure (800 psig), “HX” is a heat exchanger to preheat the reactor feed (using heat from the reactor effluent), “R” is the reactor, and “T” is a turbo-expander, to recover energy from the cooled reactor effluent. In terms of heat duty, “Q1” is the heat duty of the exchanger and “Q2” is the heat that is removed from the reactor to maintain near isothermal conditions. Note that an alternative would be to operate the reactor adiabatically and thus allow the reactor effluent to heat to a higher temperature. This would serve to increase the energy recovery from the turbo-expander. In the following discussion, CASE 1 refers to the reactor situation where the reactor effluent is predicted by constrained thermodynamics and CASE 2 refers to the case where the product selectivities are set according to the values reported in the literature.

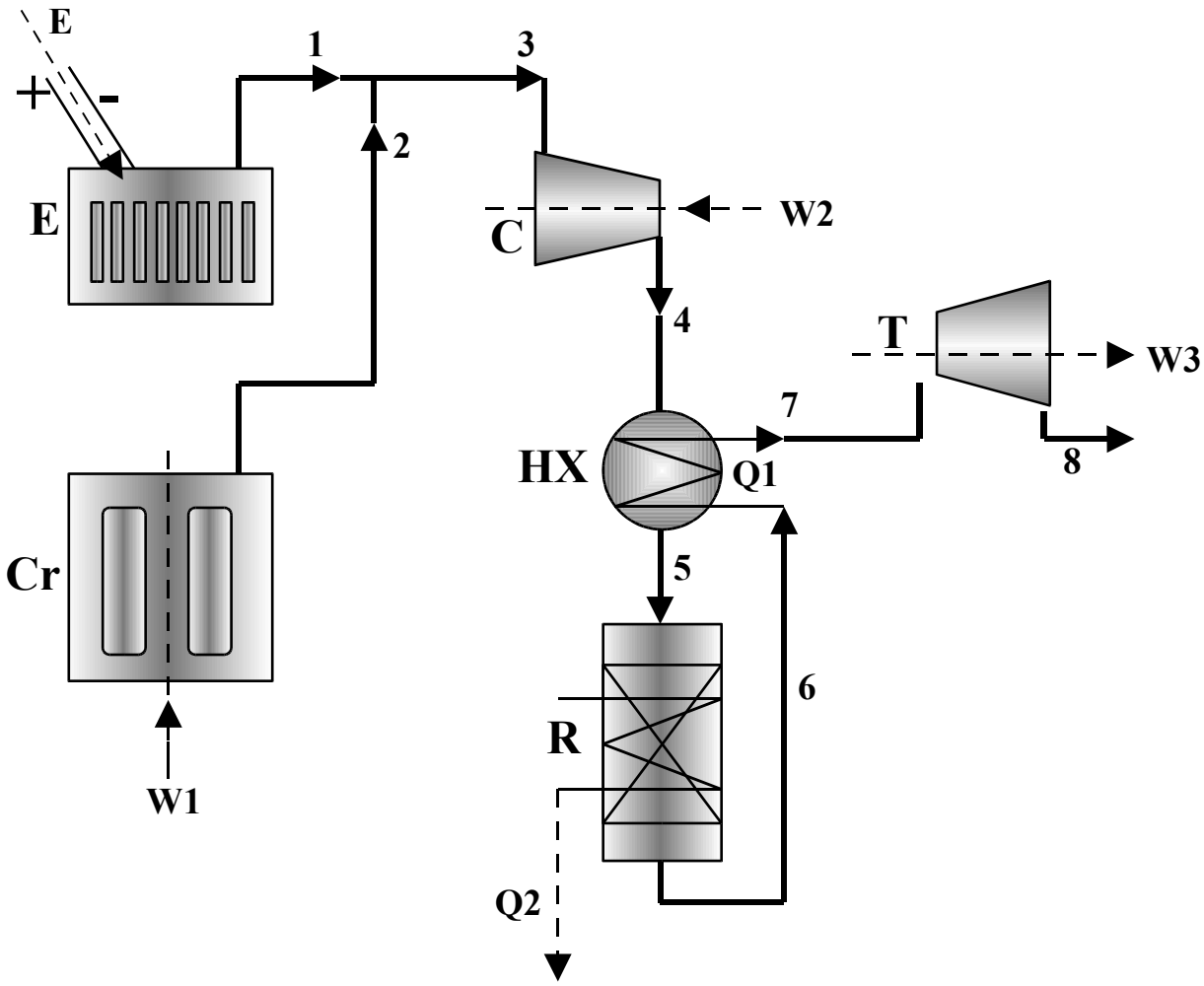
Table 1 shows the energy flows within the process for the two reactor predictions. As can be seen, the unit with the overwhelming energy consumption is the cryogenic distillation unit, consuming about 88% of the total available energy. The electrolyzer consumes about 11-15 % of the available energy. Tables 2 and 3 show the conditions and compositions of all 8 streams in the process (see Figure 1 for stream numbers) for both cases. In CASE 1, the alcohol contents are higher, whereas in CASE 2, methane supplants much of the alcohol content. However, everything else about the two cases is very similar. In terms of the heating value of the products, CASE 1 yields 270 KJ/mol and CASE 2 yields 235 KJ/mol, but CASE 2 has a 14% higher molar flow, so they both produce a possible energy rate of about 4,500 KJ/min. These values assume that the product stream (Stream 8) is available as a gas. In terms of liquid production, Case 1 would produce about 500 lbs/day of an aqueous liquid containing about 45% ethanol, whereas Case 2 would produce about 260 lb/day of an aqueous liquid containing 12.4% ethanol and 4.3% methanol.

Recommendations

The fact that the cryogenic distillation of air consumes about 88% of the available wind energy, points to the importance of that process step. It is possible that other alternatives (membrane separation, pressure swing adsorption) might prove to be a more efficient use of available energy and thus, increase the product flow rate. On the other hand, the footprint of these alternatives might be too large to be compatible with the overall design of the shipboard design. Nevertheless, this is an area that is probably in need of further exploration. The other unit that needs additional attention is the catalytic step to produce ethanol. Equilibrium favors the production of parafinnic hydrocarbons and thus the choice of the catalyst will be very important. Only a brief literature search was done and this yielded only two articles. One was in Russian and the other (referenced above) reported product selectivities that contained about 60% methane selectivity. It is recommended that a more thorough literature search be done, to determine whether a suitable catalyst does indeed exist at this point. If not, another alternative might be to produce a combustible gas (CO, CH_4, H_2) that would have a higher heating value. In other words, select a catalyst that would be more selective to these products. These catalysts are commercially available. If a liquid fuel is desired, then another alternative might be to feed the CO_2 and H_2 to a water gas shift reactor and then to a Fischer-Tropsch reactor to produce liquid hydrocarbons.

FIGURE 1

PROCESS FLOW DIAGRAM



E – Electrolyzer
Cr – Cryogenic CO_2 separator
C – Compressor
HX – Heat Exchanger
R – Reactor
T – Turbo-Expander

TABLE 1

ENERGY FLOWS – CASES 1 AND 2
Energy Flows in KW

Name	E	Q1	Q2	W1	W2	W3
CASE 1	*114.95*	2.94	5.11	*884.63*	1.16	1.91
CASE 2	*112.44*	2.91	4.68	*887.47*	1.14	2.16

TABLE 2

STREAM CONDITIONS – CASES 1 AND 2

CASE 1: Thermodynamic predictions of reaction products, assuming alcohols are the only hydrocarbons produced.

Name	1 - H2	2 - CO2	3	4
Vapor Fraction	1.00	1.00	1.00	1.00
Temperature [C]	60	25	43	139
Pressure [psig]	362	362	362	800
Molar Flow [gmol/min]	17.4	5.7	23.1	23.1
Mass Flow [g/min]	35	249	285	285
Actual Gas Flow [m3/min]	0.0188	0.0046	0.0233	0.0142
Actual Liquid Flow [cm3/min]	-	-	-	-
Heat Flow [kW]	0.3	-37.3	-37.0	-35.9
Name	**5 - RX feed**	**6 - RX prod**	**7**	**8 - product**
Vapour Fraction	1.00	1.00	0.92	0.85
Temperature [C]	368	374	189	75
Pressure [psig]	800	800	800	15
Molar Flow [gmol/min]	23.1	16.6	16.6	16.6
Mass Flow [g/min]	285	285	285	285
Actual Gas Flow [m3/min]	0.0222	0.0157	-	-
Actual Liquid Flow [cm3/min]	-	-	32.67	50.38
Heat Flow [kW]	-32.9	-38.0	-41.0	-42.9

CASE 2: Product selectivities of reaction products set to literature values

Name	1 - H2	2 - CO2	3	4
Vapor Fraction	1.00	1.00	1.00	1.00
Temperature [C]	60	25	43	139
Pressure [psig]	362	362	362	800
Molar Flow [gmole/min]	17.1	5.7	22.7	22.7
Mass Flow [g/min]	34	250	285	285
Actual Gas Flow [m3/min]	0.0183	0.0046	0.0229	0.0140
Actual Liquid Flow [cm3/min]	-	-	-	-
Heat Flow [kW]	0.3	-37.4	-37.1	-36.0
Name	**5 - RX feed**	**6 - RX prod**	**7**	**8 - product**
Vapour Fraction	1.00	1.00	0.95	0.88
Temperature [C]	368	374	171	60
Pressure [psig]	800	800	800	15
Molar Flow [gmole/min]	22.7	18.9	18.9	18.9
Mass Flow [g/min]	285	285	285	285
Actual Gas Flow [m3/min]	0.0219	0.0181	-	-
Actual Liquid Flow [cm3/min]	-	-	18.94	45.02
Heat Flow [kW]	-33.1	-37.8	-40.7	-42.8

TABLE 3

MOLAR STREAM COMPOSITIONS – CASES 1 AND 2

CASE 1 : Thermodynamic predictions of reaction products, assuming alcohols are the only hydrocarbons produced.

Name	1 - H2	2 - CO2	3	4
Hydrogen	100.00%	0.00%	75.47%	75.47%
CO2	0.00%	100.00%	24.53%	24.53%
H2O	0.00%	0.00%	0.00%	0.00%
CO	0.00%	0.00%	0.00%	0.00%
Methanol	0.00%	0.00%	0.00%	0.00%
Ethanol	0.00%	0.00%	0.00%	0.00%
Name	**5 - RX feed**	**6 - RX prod**	**7**	**8 - product**
Hydrogen	75.47%	45.28%	45.28%	45.28%
CO2	24.53%	13.38%	13.38%	13.38%
H2O	0.00%	30.36%	30.36%	30.36%
CO	0.00%	1.26%	1.26%	1.26%
Methanol	0.00%	0.02%	0.02%	0.02%
Ethanol	0.00%	9.69%	9.69%	9.69%

CASE 2: Product selectivities of reaction products set to literature values

Name	1 - H2	2 - CO2	3	4
Hydrogen	100.00%	0.00%	75.00%	75.00%
CO2	0.00%	100.00%	25.00%	25.00%
H2O	0.00%	0.00%	0.00%	0.00%
CO	0.00%	0.00%	0.00%	0.00%
Methane	0.00%	0.00%	0.00%	0.00%
Methanol	0.00%	0.00%	0.00%	0.00%
Ethanol	0.00%	0.00%	0.00%	0.00%
Name	**5 - RX feed**	**6 - RX prod**	**7**	**8 - product**
Hydrogen	75.00%	50.60%	50.60%	50.60%
CO2	25.00%	18.07%	18.07%	18.07%
H2O	0.00%	20.48%	20.48%	20.48%
CO	0.00%	1.81%	1.81%	1.81%
Methane	0.00%	7.23%	7.23%	7.23%
Methanol	0.00%	0.60%	0.60%	0.60%
Ethanol	0.00%	1.20%	1.20%	1.20%